

Functional Specification Document

1. Introduction

- **Project Title:** Hawk-Eye: 3D structure from motion solution, spatial segmentation, classification
- **Date:** 2024.10.09
- **Author:** Fady Damak
- **Approvers:**

2. Version history

#	Ver.	Date	Author	Description
00	0.0	2024.07.08.	T. Storez	Template generation
01	0.0	2024.09.25	Fady Damak	Kick-off document creation
02	0.0	2024.10.09	Fady Damak	Functional Specification

3. Purpose

- **Purpose Statement:** The purpose of this document is to outline the functional specifications for the Hawk-Eye: 3D structure-from-motion solution, focusing on spatial segmentation and classification. The purpose of this document is to outline the functionality of a system that processes synchronized video feeds to generate an interactive 3D model of a volleyball court and its elements at a user-selected frame
- **Scope:**

What is included:

- **Camera Synchronization and Recording:** Allow users to record synchronized footage from multiple camera angles.
- **Result Visualization:** Select frame using playback features to generate interactive 3D model (zoom, rotation..)
- **User Guidance:** Include step-by-step setup instructions, system checks, and documentation for using the system.

What is **excluded**:

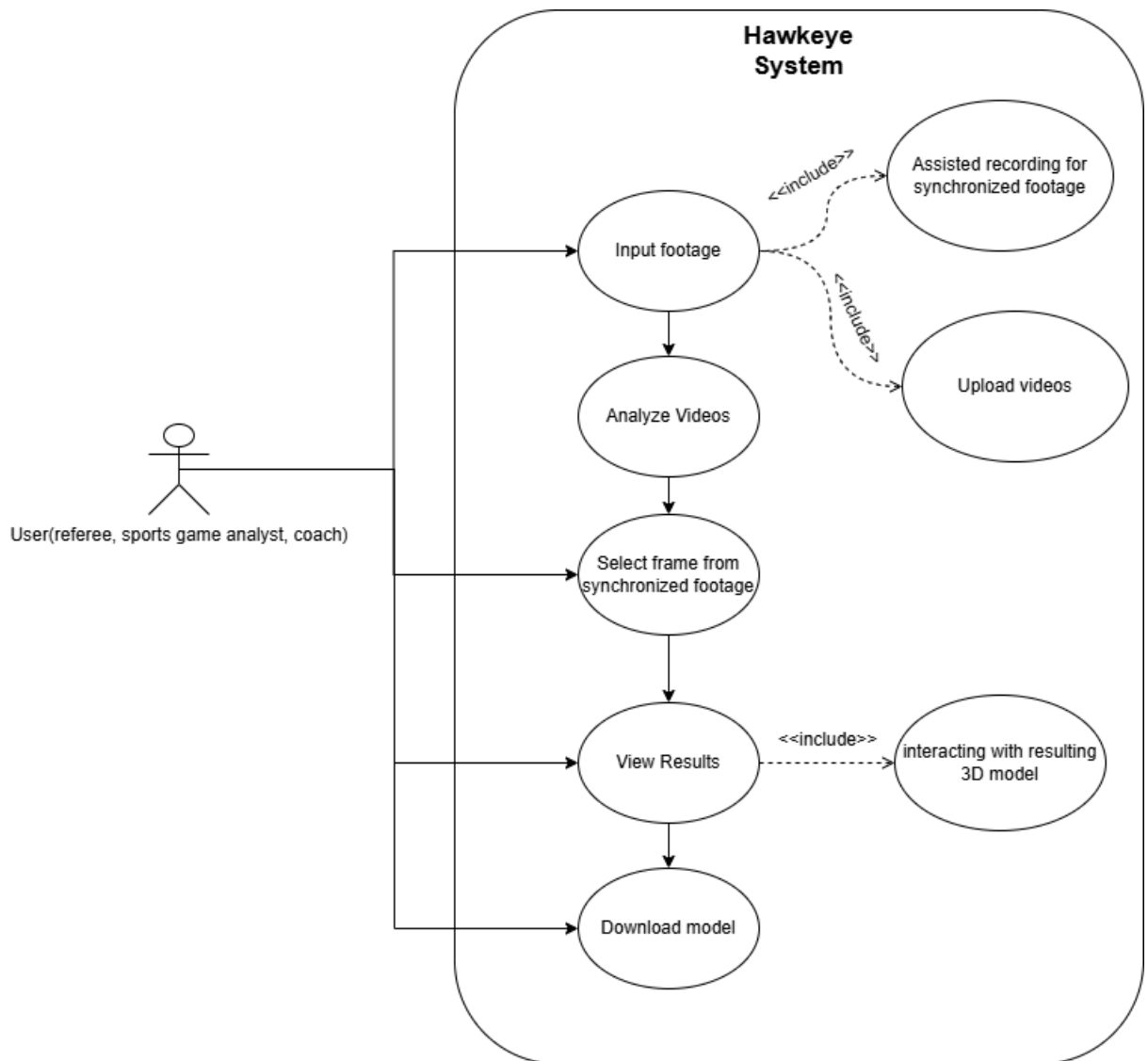
- Real-time processing or tracking during live events.

4. Overview

- **System Overview:** The **Hawk-Eye system** is designed to reconstruct the 3D trajectory of a volleyball using 2D video footage. By applying **structure from motion (SfM)** techniques, the system will analyze multiple frames to track the ball's movement, classify its position, and generate a 3D model of the selected scene then analyze ball trajectory later on. The system will support video uploads for analysis, and its primary objective is to improve sports performance analysis and refereeing decisions by providing accurate ball tracking data.
- **Assumptions:**
 - The system will initially focus solely on tracking **sports balls** (e.g., volleyball, football) rather than players or other objects.
 - The videos provided will have sufficient quality (resolution and frame rate) to allow accurate tracking of the ball's movement.
 - The court or field will be consistently visible in the video, without obstructions that would prevent clear identification of the ball.
- **Dependencies:**
 - The system is dependent on the **quality and clarity** of the video input and the **position and angle** of the cameras. Poor-quality videos may affect the accuracy of ball tracking and the 3D reconstruction process.
 - The system assumes that the videos will be recorded from **stationary cameras** to avoid complications due to camera movement.
 - The project will rely on the availability of **sport event videos** either through free online sources or future events where the system may be tested in real scenarios.
- **Technical Constraints:**
 - The system will not support real-time processing in the initial phase. Video uploads will be required, and analysis will occur after the upload.
 - The system's performance and accuracy will be influenced by the resolution and frame rate of the video.
 - The system will be limited to volleyball and tracking sports balls only in the initial phase.
- **Business Constraints:**
 - The system will be designed for sports-related use cases only, specifically for **ball tracking in volleyball (potentially other sports)**. Customizing it for other sports or objects is beyond the scope.
 - The system will not be integrated with commercial sports hardware or proprietary tracking systems but will remain software-based for video analysis.

5. Functional Requirements

- **Use Case Diagrams:**



- **Use Cases:**

Use Case 1:

- **Use Case ID:** UC-01
- **Title:** Record footage using the system
- **Description:** The user sets up the cameras and records footage of the match with assistance from the system.
- **Actors:** User (sports analyst, coach, or referee)
- **Preconditions:** The user is logged into the system; the cameras are compatible and connected to the computer.
- **Postconditions:** The video is successfully uploaded and stored for processing.
- **Triggers:** The user clicks the "Upload Video" button.
- **Main Flow:**
 - The user logs into the system.
 - The user selects the "Upload Video" option or "Record Video" option

- For the first option, the user selects a video file to upload.
- For the second option, the user is guided to record synchronized
- The system validates the video format and resolution.
- The video is uploaded and stored for further processing.
- **Alternate Flows:**
 - **AF1:** The cameras are not compatible with the software
 - **AF2:** Issues with synchronizing the videos
 - **AF3:** The user uploads the videos directly(UC-02)

Use Case 2:

- **Use Case ID:** UC-02
- **Title:** Upload Volleyball match footage
- **Description:** The user uploads a video to the system for ball tracking analysis.
- **Actors:** User (sports analyst, coach, or referee)
- **Preconditions:** The user is logged into the system; the video is in an acceptable and compatible format.
- **Postconditions:** The video is successfully uploaded and stored for processing.
- **Triggers:** The user clicks the "Upload Video" button.
- **Main Flow:**
 - The user logs into the system.
 - The user selects the "Upload Video" option or "Record Video" option
 - For the first option, the user selects a video file to upload.
 - For the second option, the user is guided to record synchronized
 - The system validates the video format and resolution.
 - The video is uploaded and stored for further processing.
- **Alternate Flows:**
 - **AF1:** The video format is invalid. The system prompts the user to upload a valid video format.
 - **AF2:** The video resolution is too low. The system notifies the user and rejects the upload.
 - **AF3:** The user records match footage using cameras with the software assistance (UC-01)

Use Case 3:

- **Use Case ID:** UC-03
- **Title:** Process Video for Ball Tracking
- **Description:** The system processes the uploaded video to track the ball and reconstruct a simple 3D model of the scene.
- **Actors:** System (automated process)
- **Preconditions:** A valid video has been uploaded to the system.
- **Postconditions:** The system generates a 3D model of the scene.
- **Triggers:** A valid video is uploaded and verified.
- **Main Flow:**
 - The system begins processing the uploaded video.
 - It extracts individual frames and identifies the ball in each frame.
 - It calculates the ball's 3D position using structure-from-motion techniques.
- **Alternate Flows:**

- **AF1:** The video quality is insufficient for accurate tracking. The system notifies the user and stops processing.
- **AF2:** Error during the video processing

Use Case 4:

- **Use Case ID:** UC-04
 - **Title:** View Ball Trajectory
 - **Description:** The user views an interactive 3D model of the scene after selecting a specific frame.
 - **Actors:** User (sports analyst, coach, or referee)
 - **Preconditions:** The footage has been synchronized
 - **Postconditions:** The user views the 3D model videos with the ability to check multiple angles and perspectives. (with playback controls and zoom)
 - **Triggers:** The user selects the "View Results" option.
 - **Main Flow:**
 - The user logs into the system.
 - The user is met with playback functionality of all the connected cameras. The footage of all cameras would be synchronized
 - The user selects a frame and an interactive 3D model of that exact moment is generated. The user can zoom in and out and rotate the model
 - The user can interact with the model or download it.
 - **Alternate Flows:**
 - **AF1:** The 3D model is not available due to errors during processing. The system notifies the user.
- **User stories:**

User Story 1:

- **Title:** Upload Sports Video/Record synchronized videos
- **As a :** As a sports analyst
- **I want to :** upload a recorded game video/ record match footage using multiple cameras
- **So that :** I can analyze the ball's movement for post-game review.
- **Format:** "As a sports analyst, I want to upload a recorded game video so that I can analyze the ball's movement for post-game review."

User Story 2:

- **Title:** View Ball position
- **As a :** As a coach
- **I want to :** view the current position of the ball at an exact moment
- **So that :** I can understand how the ball moved and make adjustments to the team's strategy.

- **Format:** "As a coach, I want to view the current position of the ball at an exact moment so that I can make strategic adjustments based on the ball's movement."

User Story 3:

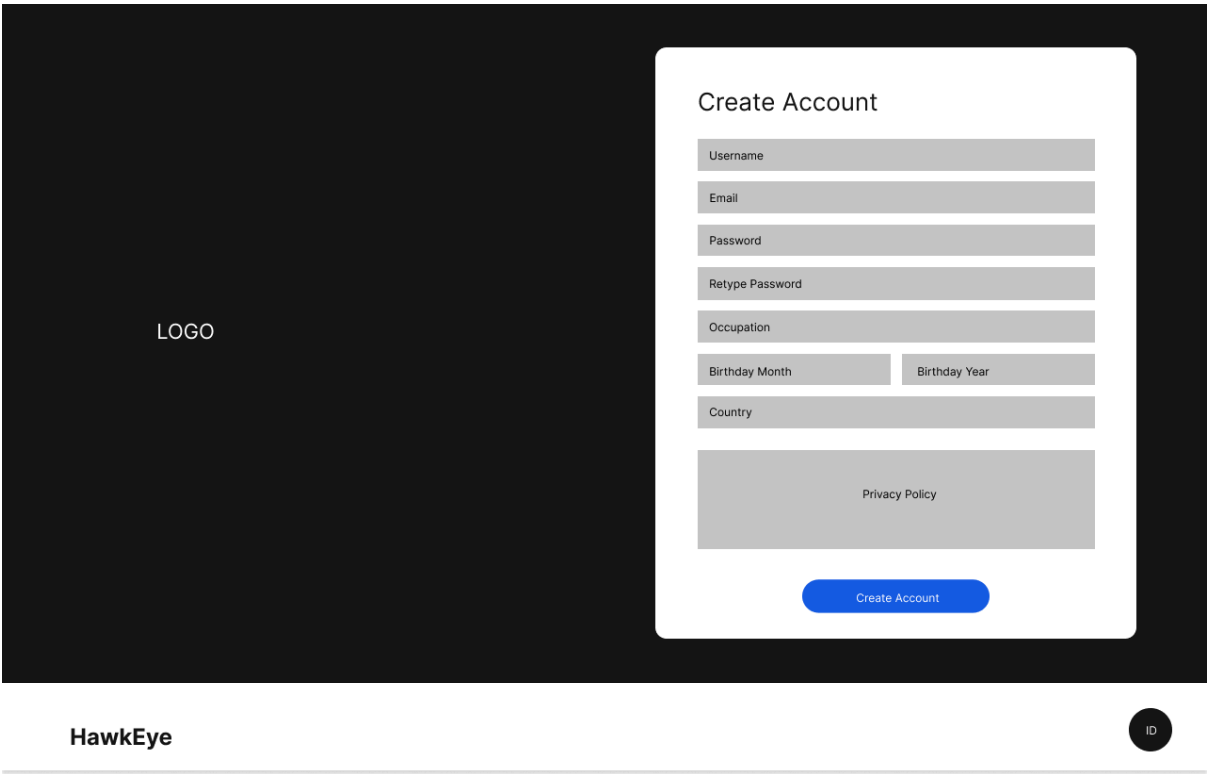
- **Title:** Track ball position
- **As a :** As a referee
- **I want to :** have the system track the ball's position
- **So that:** I can use the data to make fair, accurate game decisions.
- **Format:** "As a referee, I want the system to track the ball's movement so that I can use the data to make accurate game decisions."

6. Non-Functional Requirements

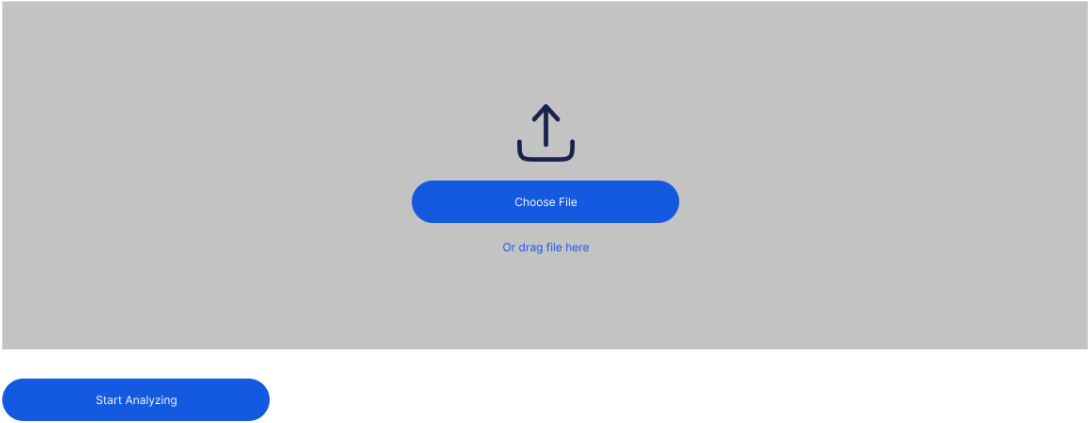
- **Performance:**
 - o The system should process uploaded videos and generate visualizations within 2-3 minutes for a typical sports video under 10 minutes with 2 cameras connected
- **Security:**
 - o Authentication is required for users to access videos uploaded and results features to safeguard each account's videos.
 - o User data and uploaded videos should be protected to prevent unauthorized access or data breaches.
- **Usability:**
 - o The hybrid viewer should support intuitive navigation, including switching views, zooming, and basic playback controls.
 - o The design should adhere to usability standards, supporting clear navigation and accessible functionality for non-technical users like coaches and analysts.
- **Reliability:**
 - o The system should be available 99% of the time, ensuring reliable access for users.
 - o Automatic error handling and recovery should be in place to manage unexpected interruptions, with notifications to users if any processing issues occur.
- **Scalability:**
 - o It should support future enhancements, such as adding player tracking or advanced analysis, without needing major system redesign making the system modular and scalable.
- **Compliance:**
 - o Any data retention policies should align with privacy laws (like GDPR), and users should have the option to delete their data upon request.

7. User Interface

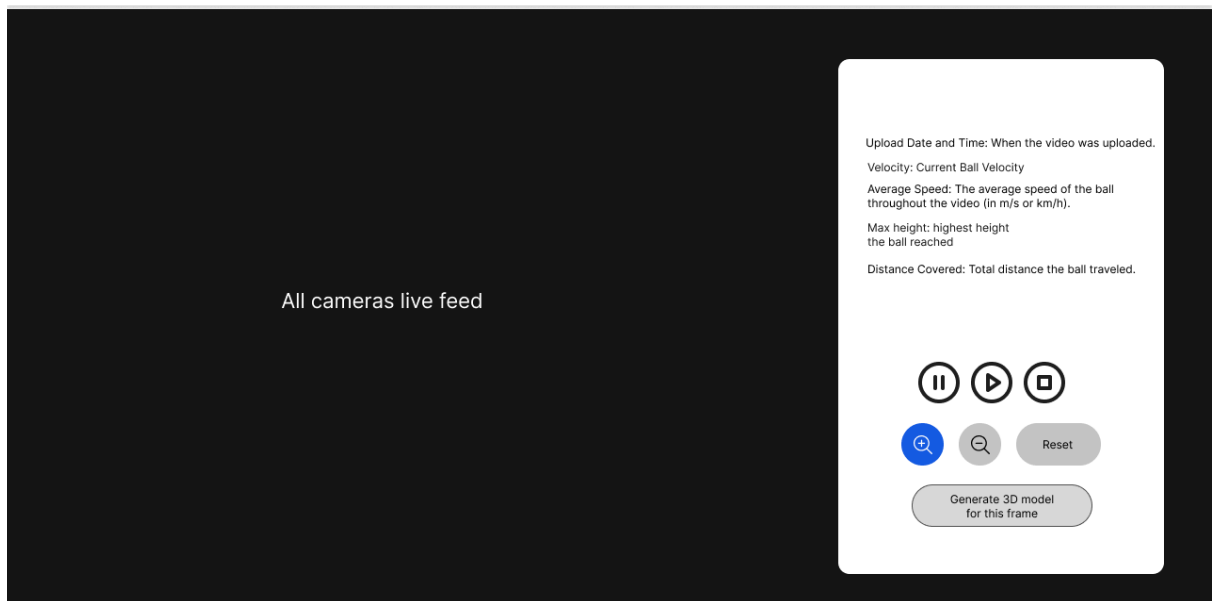
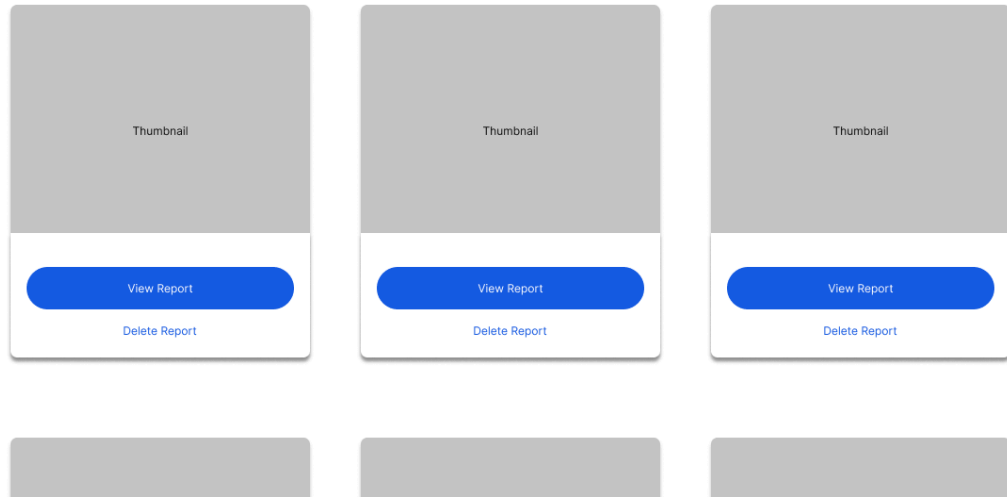
- **Wireframes/Mockups:**



Upload Video



Previous Uploads



The designs are available through this link on figma: [link](#)

- **User Interaction:**

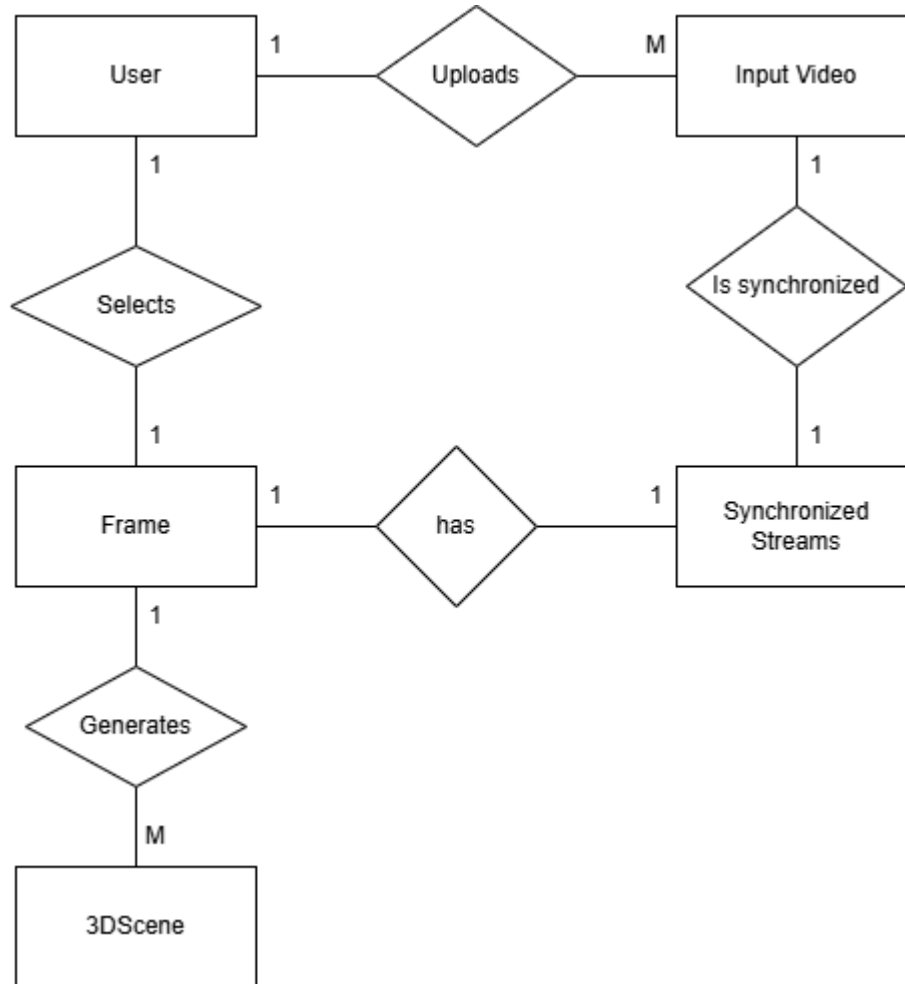
- Users can log in, upload videos, view processing status, and interact with the generated 3D trajectory output.
- User has access to all uploaded/recorded footage and he can choose which frame to generate the model.

- **Navigation:**

- The system's navigation flow will follow a linear sequence: **Upload Video/Record > Processing > View Results > Download model**.
- A navigation menu should allow users to quickly access their previous uploads and download available reports.

8. Data Requirements

- **DataModel:**



- **Data Dictionary:**

Entity	Description	Type	Constraints
Input Video	Video file uploaded/recorded by the user	MP4, AVI	Resolution: HD or higher FPS: 30 or higher
Synchronized Stream	Temporarily aligned video streams for analysis	Stream Data	Timestamps synchronized to the millisecond
Trajectory Data	3D coordinates of the ball's movement in space	Float array	Temporary, resolution dependent
Rendered Output	3D model of selected frame	3D model (obj, GLTF..)	Fixed angles(e.g. top-down, side view)

Data Flow:

Input Stage:

- User uploads a video file through the interface or records footage with assistance from the software
- The system validates the file format and size and quality of the videos.

Processing Stage:

- The uploaded video is analyzed for ball detection and trajectory extraction.
- Temporary trajectory data is generated and used to create visualizations.

Output Stage:

- The system generates 3D scene render
- These outputs are made available for the user to download.

Discard Stage:

- After the session ends, all temporary data (trajectory data, intermediate files) is deleted to maintain statelessness.

9. Integration

- **Integration Points:**
 - Libraries and frameworks will be used for the analysis and rendering of the output, present in the backend of the system.
- **APIs:**
 - **Upload API:** Enables video files to be uploaded for processing.
 - **Recording API:** Guide the user in setting up the cameras and record synchronous videos of the match to be analyzed.
 - **Processing API:** Manages the processing status and handles retrieval of trajectory data.
 - **Results API:** Provides 3D scenes for user visualization and download in supported formats.

10. Testing requirements

Testing Criteria:

- **Functional Testing:** Verify that users can upload videos, track processing status, view trajectory results, and download reports.

- **Usability Testing:** Ensure the interface is user-friendly, and typical users (e.g., coaches and analysts) can complete all tasks easily.
- **Security Testing:** Check that authentication works correctly and that unauthorized access is prevented.
- **Performance Testing:** Test the system's response time and processing speed with various video sizes to ensure it meets performance requirements.
- **Reliability Testing:** Test the system under continuous usage to verify stability and uptime.
- **Compliance Testing:** Ensure data handling complies with regulations like GDPR, and verify that users can delete their data upon request.

Testing Methodologies:

- **Unit Testing:** For individual functional components such as file uploads and data retrieval.
- **Integration Testing:** To verify that different system components, like the upload process and results retrieval, work seamlessly together.
- **User Acceptance Testing (UAT):** Have users test the system's functionality to ensure it meets their needs and expectations.